

Review Article



Pioneering pharmacometrics practice in Korea: an eight-year retrospective analysis of 192 projects from the first dedicated service organization (2016–2024)

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ABSTRACT

Pharmacometrics has become a pivotal component of drug development and regulatory science in Korea, evolving from traditional pharmacokinetic (PK)/pharmacodynamic (PD) modeling to advanced approaches such as physiologically based pharmacokinetic (PBPK) modeling and quantitative systems pharmacology (QSP). This study reviews 192 pharmacometrics projects conducted between 2016 and 2024 by Korea's first pharmacometrics service company, covering diverse modalities, indications, and development stages. Key applications included first-in-human dose prediction, clinical trial design, and patient population-specific PK/PD modeling, with systematic tracking of regulatory outcomes and publications. While oncology remained the dominant therapeutic area, recent projects expanded into gene therapy, GLP-1 agonists, pediatric orphan indications, and advanced biologics including antibody-drug conjugates, bispecific antibodies, and engineered fusion proteins—using PBPK and semi-mechanistic models. These case-driven analyses highlight the growing regulatory impact and strategic value of pharmacometrics. As the field moves into a new era shaped by artificial intelligence/machine learning integration and Food and Drug Administration-driven new approach methodologies initiatives, this review not only reflects on past practices and achievements but also provides strategic perspectives to guide the future direction of pharmacometrics in Korea.

Keywords: Pharmacokinetics; Pharmacodynamics; Clinical Pharmacology; Computer Simulation; Drug Approval

INTRODUCTION

In recent years, the U.S. Food and Drug Administration (FDA) has increasingly recognized quantitative systems pharmacology (QSP) as a key tool in model-informed drug development (MIDD), as evidenced by the establishment of the FDA's Quantitative Medicine Center of

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Excellence [1]. This represents a broader regulatory shift toward incorporating quantitative modeling approaches to support drug approvals and clinical decision-making. As global regulatory agencies adopt model-based strategies, integrating mechanistic and empirical models has become essential for optimizing drug development and improving clinical outcomes globally [2-4].

Pharmacometrics has traditionally relied on empirical pharmacokinetic (PK)/pharmacodynamic (PD) modeling using top-down approaches to describe observed data and optimize dosing regimens. However, recent advances in computational tools and data availability have highlighted the advantages of integrating mechanistic, systems biology-based methods. The convergence of population PK (PopPK)/PD (top-down) and mechanistic modeling (bottom-up) now enables more accurate dose-exposure-response predictions, especially in complex therapeutic areas [3,4]. This integrative approach offers a more holistic understanding of drug behavior in both biologic and physiologic systems, incorporating molecular-level physiology and drug-specific mechanisms of action to ultimately benefit patients [5].

In alignment with these global streams, Korea has taken important steps to harmonize its regulatory framework with international standards. The Ministry of Food and Drug Safety (MFDS) issued its first modeling-related guideline in 2008 on the design and evaluation of dose-response studies, followed by PopPK modeling guidelines and PK/PD-based dose setting for antibiotics in 2015 and 2016, respectively [6,7]. These efforts have accelerated the adoption of model-informed strategies in both regulatory review and drug development in Korea. Over time, we observed more complex mechanistic modeling approaches gradually integrated into academic and industrial settings for pharmacometrics practice.

The application of pharmacometrics has expanded rapidly in parallel with the growth of biologics and advanced therapeutic modalities [8-10]. In Korea, several studies have addressed different aspects of the pharmacometrics landscape, including regulatory perspectives on drug approval processes, historical analyses of clinical pharmacology development, and early summaries of pharmacometrics practice based on service organizations [11-13]. Global reviews have also highlighted the expanding role of regional pharmacometrics and MIDD in modern drug development and pharmacotherapy [14]. However, published analyses describing regional pharmacometrics ecosystems remain limited, particularly those integrating large number of real-world projects across therapeutic areas and development stages.

A major milestone was reached in 2016 with the establishment of Korea's first pharmacometrics service company, marking the transition from academia-driven modeling to industry-level pharmacometrics services supporting regulatory submissions and out-licensing for regional pharmaceutical and biotech companies [11]. In prior study, our group summarized the first 3 years of this experience from this initial pharmacometrics service organization in Korea [11].

Over the past decade, pharmacometrics in Korea has evolved from a supporting tool to a central component of clinical development. Recognized by MFDS for its ability to optimize dosing, improve trial design, and support both regional and international regulatory submissions, pharmacometrics has become increasingly embedded in Korea's drug development ecosystem [14,15].

Building on this prior work, the present study provides an updated and expanded analysis incorporating more recent projects, offering a cumulative perspective on the evolving

landscape of pharmacometrics practices in the Korean drug development ecosystem. In this study, we systematically evaluate 192 pharmacometrics project experiences in Korea from 2016 to 2024, spanning various modalities and therapeutic areas. While traditional PK/PD approaches remain foundational, there is a clear trend toward incorporating mechanistic models. By characterizing key projects and presenting lessons learned by case studies, we highlight the practical application of modeling based on nonclinical and clinical data. This review provides an overview of evolving trends in Korean pharmacometrics and its broader impact on drug development, while also proposing future directions for the field in the context of artificial intelligence (AI)-driven drug development.

MATERIALS AND METHODS

The project data were collected over a period of 100 months (~8 years) from 2016 to 2024 and categorized based on several key attributes, including drug indication, modality, therapeutic area, stage of service, project type, detailed project specifications, final objectives, journal publication details, sequential project information to track revisited clients, and the achievements and outcomes derived from the pharmacometrics analysis results. The collected data were converted into numeric categorical variables, then statistically summarized and visualized using various tables and plots. Data extraction and all graphical analyses were performed using Python (version 3.9) with pandas and matplotlib libraries.

OVERVIEW OF COMPREHENSIVE PHARMACOMETRICS PROJECTS EXPERIENCES

Number of projects and companies

Over the past 8 years, a total of 192 projects were completed, counting only main projects with clearly defined objectives and excluding duplicates resulting from contractual transfers. These projects were commissioned by 75 different companies, with an average of approximately 2.5 projects per company. Notably, 45 companies (60% of the client pool) requested 2 or more projects with a high level of client retention observed in Korea-based industry.

The client pool who needed pharmacometrics analysis was classified into 5 categories: Regional (Korea-based) (1) Pharma, (2) Biotech, (3) Contract Research Organizations (CROs), (4) Foreign companies (U.S. biotech), and (5) Other. Among these, biotech companies represented the largest share at 50%, followed by Pharma at 37%, Other at 7%, CROs at 5%, and U.S. biotech companies at 2% (**Fig. 1**). Pharma clients included 7–8 of the top 10 highest-selling pharmaceutical companies in Korea, as well as other major bio-tech companies. CROs primarily acted as intermediaries, outsourcing modeling work on behalf of regional pharmaceutical companies. U.S. biotech clients were developed through Korea–U.S. collaborative business models, while the Other category included consulting firms subcontracting pharmacometrics analyses.

The cumulative number of projects steadily increased, reaching 192 by 2024, starting from 4 projects in the initial year, 2016. Annual project counts varied, with a peak of 35 projects in 2022. The most rapid growth in cumulative projects occurred between 2019 and 2023, averaging 30.5 new projects per year during that period. Annual project numbers rose significantly until 2020, remained relatively stable between 2020 and 2022 with an average

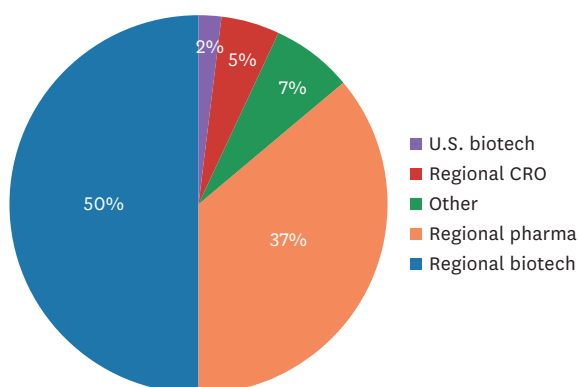


Figure 1. Distribution of client company types utilizing pharmacometrics services. CRO, Contract Research Organization.

of 32.7 projects per year, and showed a modest decline in 2023 and 2024, during which 23 projects were conducted each year. Overall, the trend demonstrates sustained growth in pharmacometrics project activity, reflecting ongoing demand for modeling and simulation (MS) services in Korea over the past decade (**Fig. 2**).

Scope of practices

The distribution of project scopes from 2016 to the present is illustrated in **Fig. 3**. Projects were categorized into 5 types: 1) MS—including PopPK, PBPK, and semi-mechanistic models; 2) Drug-drug interaction (DDI) prediction—PBPK models developed specifically for prediction of dynamic time-course of DDIs; 3) Clinical Pharmacology consulting (Consult) involving interpretation, extension, and application of pharmacometrics modeling results to practical clinical development decisions; and 4) other tasks, such as report writing and supplementary output generation (Other). Categories 1–4 primarily involved pharmacometrics modeling as a major analytical component. The fifth category was non-compartmental analysis (NCA), which represents a traditional PK analysis method that is model-independent.

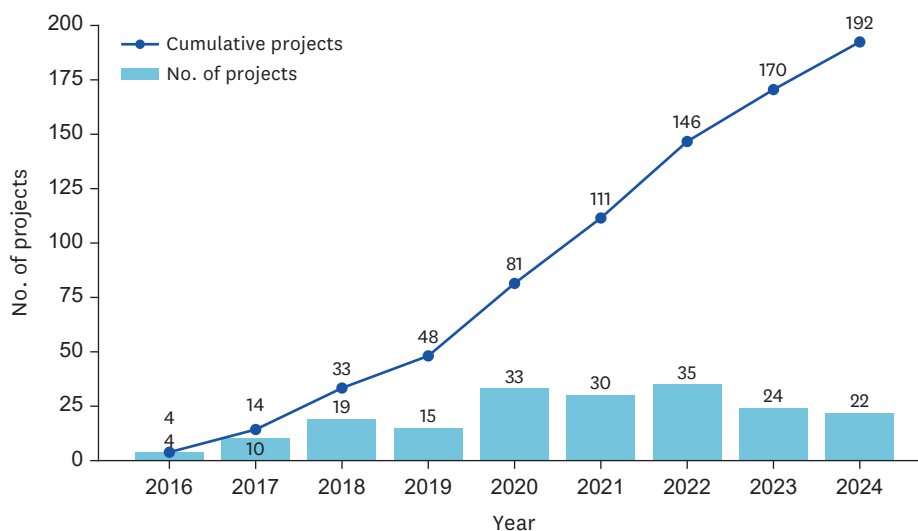


Figure 2. Number of pharmacometrics projects per year.

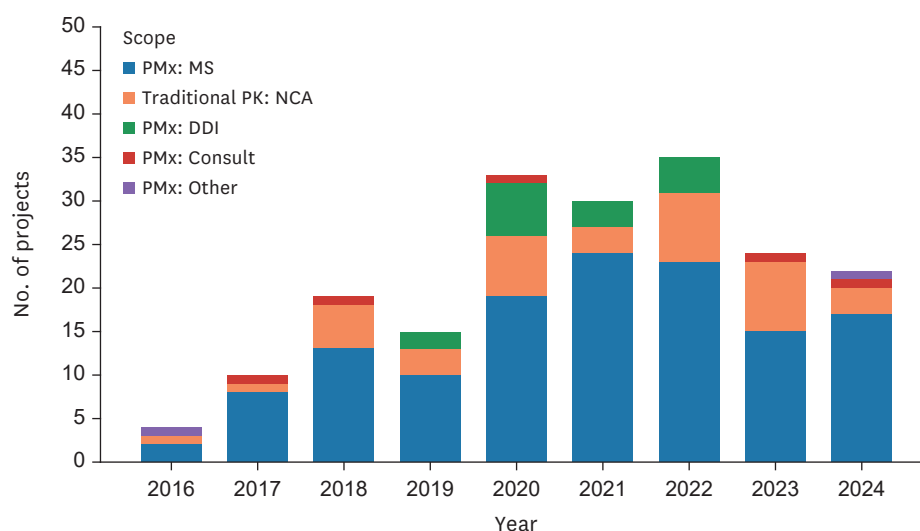


Figure 3. Pharmacometrics projects per year per scope.

PMx, pharmacometrics modeling analysis as a major analysis; MS, modeling and simulation; Traditional PK, model-independent traditional pharmacokinetic analysis; NCA, non-compartmental analysis; DDI, drug-drug interaction.

MS consistently accounted for the largest proportion of projects each year, with notable peaks in overall project numbers in 2020 and 2022, corresponding to increased demand and improved workflow efficiency supported by expanded pharmacometrician capacity. Contributions from other scopes, including NCA and DDI, also increased in proportion to the overall growth in annual project volume.

Indication and therapeutic areas

A total of 15 therapeutic areas and 42 indications were covered in the pharmacometrics projects included in this analysis. The leading therapeutic areas were Oncology, Gastroenterology, Rare Diseases, Endocrinology, Neurology and Immunology. Oncology represented the largest share, accounting for 46.35% of all projects (89 out of 192), with solid tumors being the most common indication, followed by leukemia, lymphoma, and non-small cell lung cancer (NSCLC). These Oncology modeling efforts support the goals of Project Optimus by enabling efficient dose optimization using integrated nonclinical and clinical data early in development programs, through exposure–response characterization and simulation-based evaluation of various dosing scenarios, to guide dose finding that maximizes not only efficacy but also safety and tolerability in oncology drug development [16]. Gastroenterology (8.85%) primarily involved projects related to metabolic dysfunction-associated steatohepatitis (MASH), while Rare Diseases (8.33%) focused on hemophilia and pediatric orphan indications. Endocrinology (6.77%) included an emerging trend in type 2 diabetes and obesity therapeutics, reflecting areas of growing interest in recent drug development. Other areas, such as Neurology, Immunology, Infectious Diseases, and Hematology, accounted for smaller proportions, illustrating the broad applicability of pharmacometrics analysis, as shown in **Figs. 4 and 5**.

At the indication level, the most frequently studied conditions included solid tumors, MASH, hemophilia, NSCLC, leukemia and lymphoma, Alzheimer's disease, type 2 diabetes mellitus, obesity, COVID-19 treatment, and viral infections—ranking among the top ten indications for which pharmacometrics analyses and consultations were requested by the regional pharmaceutical industry.

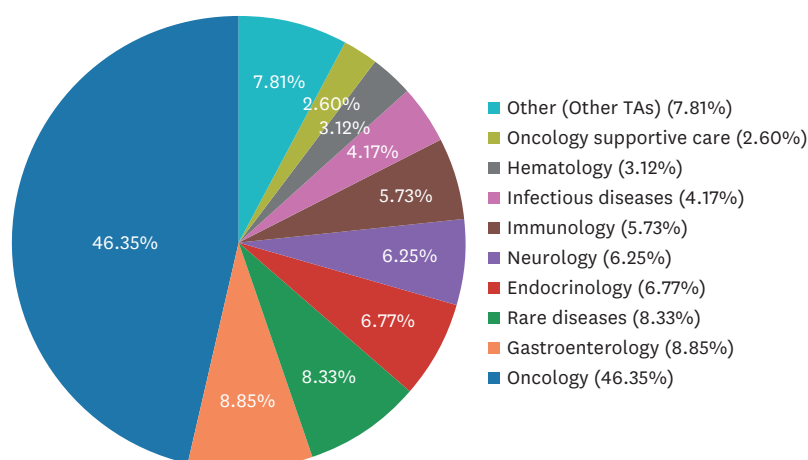


Figure 4. Distribution of TAs.
TA, therapeutic area.

In line with global trends, the regional pharmacometrics field has also reflected the rise of emerging therapeutic modalities within Korea's drug development programs. This includes efforts to enhance efficacy and optimize existing drug platforms, as well as the creation of novel combination forms such as bispecific antibodies (BsAbs) and antibody-drug conjugates (ADCs) [17-22]. A selected list of pharmacometrics modeling projects involving emerging modalities—including BsAbs, gene therapies, ADCs, and therapeutic proteins—utilizing

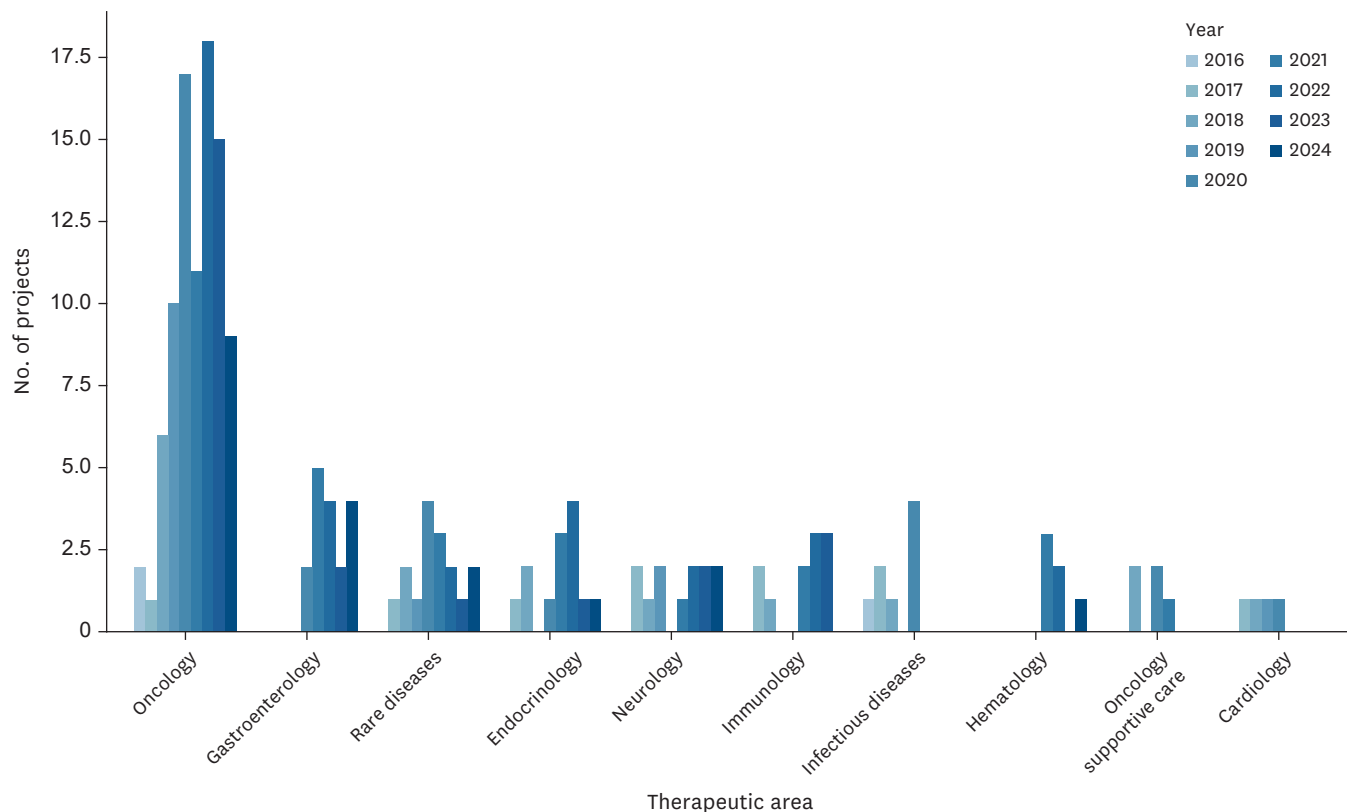


Figure 5. Yearly trends in therapeutic areas of pharmacometrics services (2016–2024): annual project counts by the top 10 therapeutic areas.

Table 1. Key features of pharmacometrics project cases involving emerging therapeutics using advanced modeling approaches (selected)

Case study (category)	Year	Company	Therapeutic area	Indication	Modality	Scope of services/Models & key considerations	Final goal achieved
1	2021	Pharma	Oncology	Advanced solid tumor	BsAb	- FIH starting dose prediction using monkey data - ADA effect analysis	MFDS Phase 1 IND became effective
2	2023	Biotech	Oncology	Solid tumor	mAb (antibody conjugate)	- PK/PD modeling using Phase 1, 2 studies - Multiple analyte analysis	Proceeded to a Phase 2 study following IND amendment
3	2022	Biotech	Endocrinology	Obesity	Fc fusion protein	- PK-food intake-body weight modeling using monkey data - FIH dose prediction	FDA Phase 1 IND became effective
4	2024	Pharma	Ophthalmology	Neovascular age-related macular degeneration	Gene therapy	- Efficacious dose prediction using <i>in vivo</i> data - Multiple analytes from multiple intraocular sites	Currently under development
5	2024	Biotech	Oncology	Advanced solid tumor	ADC	- Semi-mechanistic model development - FIH dose prediction	Currently under development
6	2024	Biotech	Oncology	Advanced solid tumor	BsAb	- Semi-mechanistic model for dual target - Efficacious dose prediction for Phase 2	Currently in Phase 1 study

BsAb, bispecific antibody; FIH, first-in-human; ADA, anti-drug antibody; MFDS, Ministry of Food and Drug Safety; IND, Investigational New Drug; mAb, monoclonal antibody; PK, pharmacokinetic; PD, pharmacodynamic; FDA, Food and Drug Administration; ADC, antibody-drug conjugate.

advanced or hybrid (mechanistic + empirical) modeling approaches is summarized in **Table 1**. These case studies illustrate diverse and in depth modeling scopes—such as first-in-human (FIH) dose and efficacious dose prediction using semi-mechanistic model development, and multi-analyte analysis—conducted to support regulatory submissions or guide dose selection for internal decisions in early-phase clinical development.

Stages in drug development

Projects supported all stages of the drug development cycle, with a stronger focus on early clinical trials and the transition from nonclinical studies.

MS projects were predominantly conducted at the Nonclinical (55.73%) and Clinical (43.51%) stages, particularly in Phase 1 (comprising 33.59% of total MS projects). These efforts focused on FIH safe dose prediction, efficacious dose prediction, optimization of dosing regimens, sampling strategies, and covariate analysis, with increasing emphasis in recent years on modeling to support the transition from Phase 1 to Phase 2. Projects involving Phase 3 data also integrated information from earlier clinical stages to develop comprehensive models supporting regulatory submissions (e.g., New Drug Application [NDA]/Biologics License Application [BLA] to the FDA or European Medicines Agency [EMA]), providing detailed characterization of the drug's PK and PD profiles in target populations (**Fig. 6**).

DDI projects were most commonly conducted during Phase 1 (68.75%), followed by the Nonclinical stage (18.75%), using basic static, dynamic static, and PBPK models to evaluate DDI potential early in development. NCA projects were based entirely on clinical data and contributed to clinical study reports, ensuring regulatory requirements were met for PK and PD data analysis (**Fig. 6**).

Consultation projects were primarily focused on early development, with the majority conducted in Phase 1 (64.29%) and Nonclinical stages (28.57%). These included manuscript preparation, interpretation of modeling results, clinical pharmacology guidance for drug repurposing, and support for regulatory responses to facilitate progression within the development program (**Fig. 6**).

These analyses provided practical insights into the predictability and utility of model-based

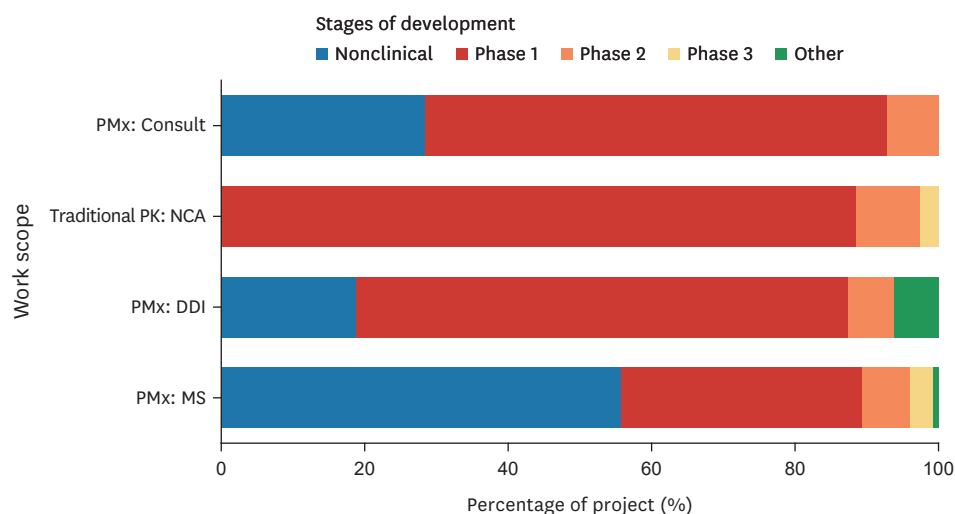


Figure 6. Stacked bar chart of distribution of projects across drug development stages.

PMx, pharmacometrics modeling analysis as a major analysis; Traditional PK, model-independent traditional pharmacokinetic analysis; NCA, non-compartmental analysis; DDI, drug-drug interaction; MS, modeling and simulation.

dose projection from nonclinical data. In most nonclinical-based modeling projects, PK data from multiple animal species (typically 2–4 species) were analyzed using compartmental models, followed by allometric scaling to extrapolate human PK parameters. For small molecules, this approach was mainly used to estimate a potential efficacious dose range while ensuring projected exposures remained below No Observed Adverse Effect Level-derived safety margins. For biologics, PK models were commonly developed using monkey data, and human PK parameters were predicted using one-species allometry with fixed exponents. Across multiple projects, monkey-based PK modeling demonstrated strong translational predictability for human exposure for biologics, while small-molecule efficacious dose predictions were more reliable when supported by reference data such as comparator drug information from structurally or mechanistically related compounds.

Although physiologically based pharmacokinetic (PBPK) models may provide a more mechanistic framework for describing interspecies differences, nonclinical analyses in practice often rely on compartmental or population-based approaches due to the limited availability of species-specific physiological parameters and relevant *in vivo* data required for PBPK model development. Clinical utility was high in case where interspecies differences were minimal, or exclusion of the most deviating species from the dataset was considered with appropriate scientific justification. However, for novel compounds mechanistic extrapolation incorporating drug-specific mechanisms of action and species-specific PBPK models is expected to improve translational prediction outcomes. With recent advances in computational methods, AI-assisted modeling, and expanded physiological data resources and collection tools, the feasibility of developing species-specific PBPK and mechanistic model libraries is expected to increase in modeling practices, even at smaller research group and individual investigator levels.

Modeling based on clinical datasets was predominantly performed using compartmental PopPK approaches combined with covariate analysis to identify key covariates mainly affecting PK parameters and drug exposures. These models were primarily applied to support dose optimization in target patient populations and to inform optimal clinical trial

design. Compared with nonclinical modeling, clinical data-based analyses typically involved more extensive simulation exercises to characterize interindividual variability, predict exposure–response relationships, and evaluate potential safety risks. Consequently, clinical pharmacometrics analyses require greater emphasis on variability modeling and predictive simulations to ensure that dosing strategies are appropriate for majority of patients in the target population. In addition, recent trends and inquiries increasingly involve the development of mechanism-based QSP models to characterize PD responses.

Final goals of the pharmacometrics projects

The majority of pharmacometrics modeling projects were conducted to support regulatory submissions to the FDA, EMA, and MFDS for Investigational New Drug (IND), NDA, or BLA applications. Submissions to other regulatory agencies were also observed, while the number of projects for internal decision-making has shown a slight increase in recent years—particularly to support the efficient development of complex modalities.

Over time, the number of projects by purpose displays a bell-shaped distribution across categories, with slight decreases in absolute submission-support project counts in recent years (**Fig. 7**). However, the complexity and sophistication of models—particularly mechanistic approaches designed to represent systems biology—have reached unprecedented levels. This reflects a shift from descriptive models toward mechanistic, systems-based strategies that better address the complexity of novel modalities.

Approximately 75% of pharmacometrics analysis projects contributed directly to regulatory submissions or provided supporting evidence for submission documents (**Fig. 8**). The most frequent submissions were made to Korea's MFDS, followed by the FDA and EMA, primarily for IND and NDA/BLA applications. Additional regulatory agencies included Federal Institute for Drugs and Medical Devices (BfArM, Germany), Medicines and Healthcare Products Regulatory Agency (MHRA, UK), and agencies in China, Australia, and Taiwan—mostly for Phase 1 IND rationale or NDA supplements.

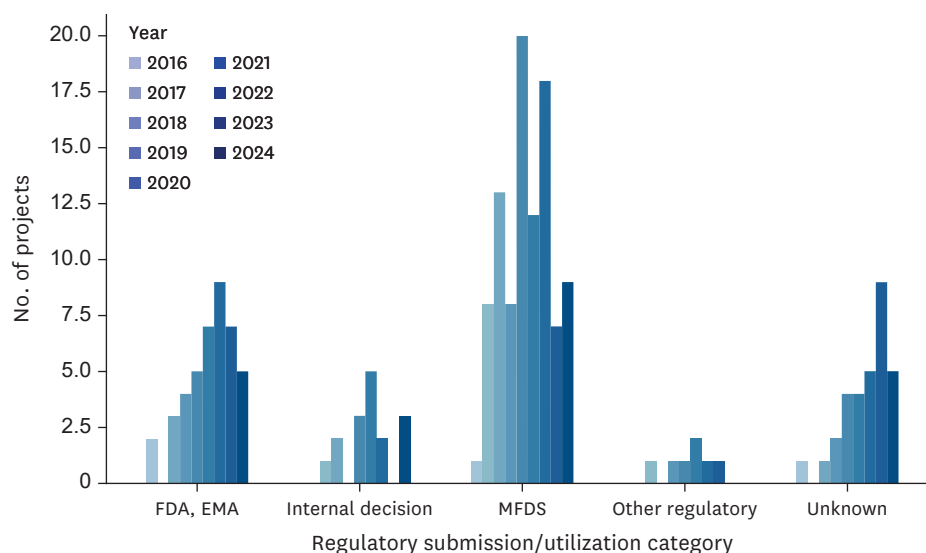


Figure 7. Annual number of projects by regulatory submission and utilization category. FDA, Food and Drug Administration; EMA, European Medicines Agency; MFDS, Ministry of Food and Drug Safety.

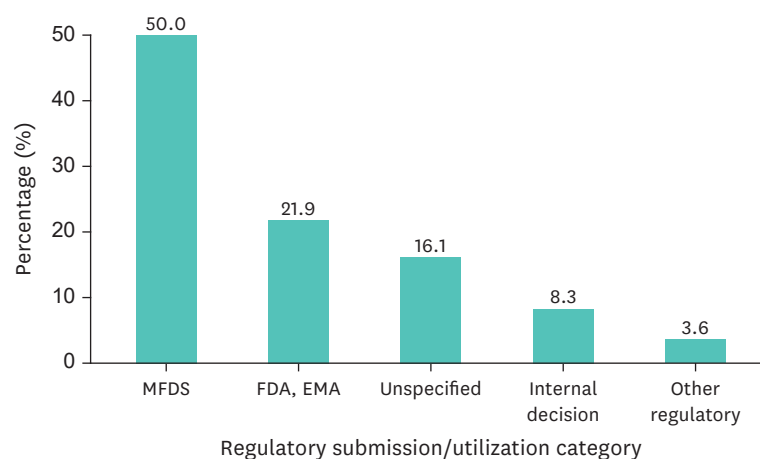


Figure 8. Percentage distribution of projects by regulatory submission and utilization category. MFDS, Ministry of Food and Drug Safety; FDA, Food and Drug Administration; EMA, European Medicines Agency.

Submissions to MFDS often involved Phase 1, 1b, and 2 IND applications, with pharmacometrics analysis used to justify optimal dosing regimens. In some cases, model-based evidence helped obtain conditional approvals or waivers by demonstrating low safety risks in vulnerable populations such as pediatric patients or those with comorbidities.

To further characterize the regulatory context of these submissions, the distribution of pharmacometrics analysis scope across regulatory agencies was evaluated (**Fig. 9**). Projects submitted to MFDS most frequently focused on IND-enabling dose projection and early clinical trial design support, reflecting the emphasis on FIH dose justification and initial clinical development planning. In contrast, submissions involving FDA and EMA more often included dose-finding, dose optimization, and PopPK analyses including covariate analysis to support later-stage development decisions. These differences highlight how pharmacometrics applications are tailored to the regulatory requirements and development stages associated with each agency.

Overall, the number of regulatory-support projects has increased in parallel with the total number of annual projects, highlighting the growing integration of pharmacometrics analyses into both regional and global regulatory strategies, as well as the increasing acceptance of model-based approaches by regional regulatory authorities.

Achievements/outcomes of good pharmacometrics practices

Publishing pharmacometrics analyses in peer-reviewed journals is recognized as good practice, and **Table 2** outlines selected examples that meet this standard. Although hesitancy remains in regional industry due to concerns about sharing raw data and early-stage efficacy results, several forward-looking projects have successfully progressed to publication. These works reflect high-quality pharmacometrics analyses that not only supported regulatory approvals but also provided long-term strategic values to companies and researchers.

PHARMACOMETRICS ANALYSIS CASE STUDIES

We present 5 advanced case studies illustrating the application of pharmacometrics modeling across various contexts, including pediatric dose selection for an orphan drug, translational

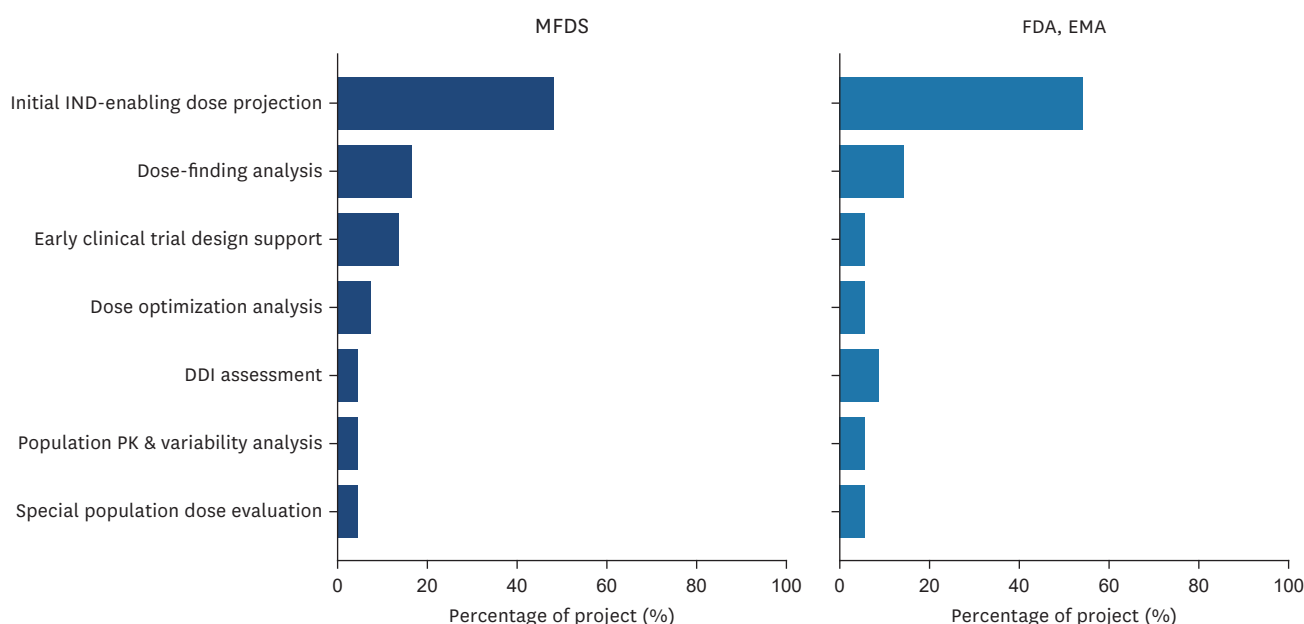


Figure 9. Distribution of detailed pharmacometrics service scope in regulatory submissions by agency. MFDS, Ministry of Food and Drug Safety; FDA, Food and Drug Administration; EMA, European Medicines Agency; IND, Investigational New Drug; DDI, drug-drug interaction; PK, pharmacokinetic.

Table 2. Published research articles from completed pharmacometrics modeling projects (selected)

Case study (year)	Company	Indication	Therapeutic area	Stage	Scope of services	Publication information	Final goal achieved
1 (2017)	Daewoong Pharmaceutical	Type 2 diabetes mellitus	Endocrinology	Non-clinical	- PK/PD model development using <i>in vivo</i> PK and cUGE data - Prediction of efficacious dose and human PK/PD	<i>BJCP</i> (DOI: 10.1111/bcp.15891) [23]	MFDS Phase 1 IND became effective
2 (2021)	GC Biopharma Corp.	Hemophilia	Rare diseases	Non-clinical	- TMDD model development using monkey data - FIH dose prediction for mAb	<i>J Thromb Haemost</i> (DOI: 10.1111/jth.15244) [24]	MFDS Phase 1a IND became effective
3 (2020)	Cumine	-	-	-	- PBPK modeling of biocides for risk assessment	<i>Arch Toxicol</i> (DOI: 10.1007/s00204-021-03041-z) [25]	MOE modeling project
4 (2021)	Samsung Bioepis	Paroxysmal nocturnal hemoglobinuria	Immunology	Phase 1/3	- PK/PD-Response model development using all clinical data - Covariate analysis - Assess impact of race	<i>Eur J Clin Pharmacol</i> (DOI: 10.1007/s00228-024-03703-8) [26]	FDA BLA approval, MFDS, EMA drug approval, NMPA (China) supplemental document submission
5 (2023)	SN BioScience	Solid cancer	Oncology	Phase 1	- Multi-compartment PK model development - PK and multiple analyte analysis	<i>Frontiers</i> (DOI: 10.3389/fphar.2025.1556618) [27]	FDA Phase 1 IND became effective

PK, pharmacokinetic; PD, pharmacodynamic; cUGE, cumulative urinary glucose excretion; MFDS, Ministry of Food and Drug Safety; IND, Investigational New Drug; TMDD, Target-Mediated Drug Disposition; FIH, first-in-human; mAb, monoclonal antibody; PBPK, physiologically based pharmacokinetic; MOE, Ministry of Environment; FDA, Food and Drug Administration; BLA, Biologics License Application; EMA, European Medicines Agency; NMPA, National Medical Products Administration.

PK/PD modeling from nonclinical to clinical phases, modeling of a GLP-1 agonist for obesity, PBPK-based DDI prediction, and specialized formulation modeling for AI-discovered drug combinations. These cases demonstrate how pharmacometrics analyses support clinical decision-making, patient-specific dosing, and translational modeling from preclinical data to human application, ultimately contributing to impactful regulatory decisions in the drug development program.

For case studies, figures are presented with predicted and simulated values shown as relative or scaled values, or with dose levels masked as necessary to comply with confidentiality

constraints for each project, while retaining the original model structure and parameter estimates, in order to show the analytical workflow and the form in which the modeling results were generated.

Case study 1. Prediction of pediatric dosing for rare diseases using PopPK modeling and virtual patient simulations to identify the maximally effective dose

Goal: To predict pediatric dosing for a rare orphan drug and support FDA clearance of the IND amendment to initiate a Phase 2 pediatric clinical study.

Key point: PopPK modeling was essential for predicting an optimal dose that would achieve effective drug exposure across pediatric age groups while maintaining acceptable safety margins. In this case, the FDA specifically emphasized the importance of selecting a dose expected to achieve exposure levels exceeding the predefined efficacy threshold, and the model-informed analysis was used to demonstrate that the proposed regimen met this regulatory expectation for efficacy and safety.

Methods & results: A PK model was developed using Phase 1 data from healthy adult subjects, incorporating covariates such as body weight, formulation, food status, and special PK groups. Body weight was implemented as a mechanistic covariate, and allometric scaling was applied to extrapolate PK profiles across pediatric population. Virtual patient simulations were conducted, reflecting realistic distributions of weight and age of the target patient population, to identify optimal weight-based dosing regimens and to determine whether predicted exposures would reach the predefined efficacy threshold derived from nonclinical *in vivo* efficacy data.

The predefined efficacy threshold was derived from *in vivo* efficacy studies and adult PK data from the Phase 1 Single Ascending Dose study were used to illustrate the range of tolerated exposures observed in humans. We have already observed that in a subset of healthy adult subjects with relatively high PK variability, substantially higher exposures were well tolerated without safety concerns. These observations provided an additional safety margin for evaluating the proposed pediatric dosing regimen.

The simulation results indicated that the proposed regimen would produce pediatric exposures exceeding the efficacy threshold across the majority of simulated patients, while remaining within the tolerated exposure range observed in adults. The figures show predicted drug exposure at the highest proposed pediatric dose cohort, demonstrating exposures above the efficacy threshold, while remaining within the observed safety window (**Figs. 10 and 11**).

Outcomes: Based on FDA feedback, pediatric dosing cohorts recommended for inclusion in the Phase 2 study were those predicted to achieve or exceed the efficacy threshold as proposed, while dosing regimens that did not meet this criterion were not pursued, highlighting the FDA's emphasis on achieving adequate efficacy in this pediatric orphan drug development case. The modeling-informed dosing strategy supported FDA clearance of the IND amendment to proceed with the Phase 2 pediatric study. The Phase 2 trial is currently ongoing.

Case study 2. PK/PD modeling from non-clinical prediction to Phase 1 efficacy and Phase 2 guidance

Goal: To utilize PK/PD MS to bridge non-clinical data with clinical dose selection, predict the efficacious dose for Phase 1 trials, establish a safe and effective starting dose using the

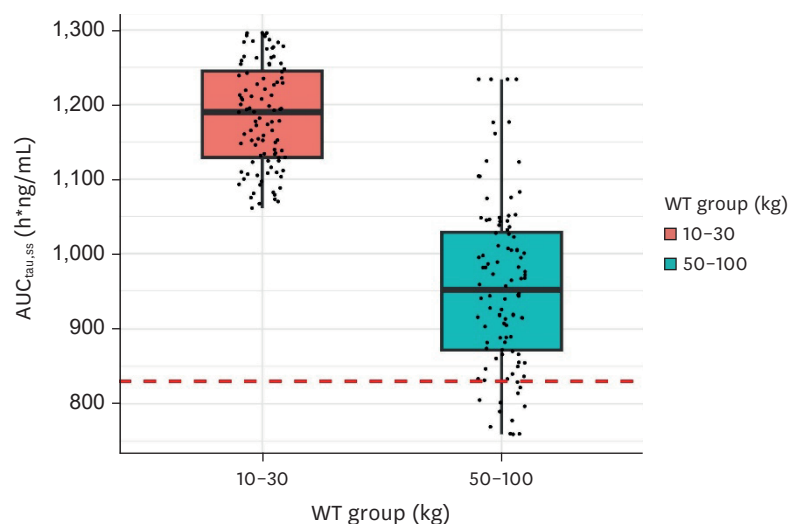


Figure 10. Simulated $AUC_{\tau,ss}$ predictions comparing pediatric patients and healthy adults under the proposed dosing regimen (illustrative). Red-dashed line represents target efficacy threshold derived from *in vivo* studies. AUC, area under the curve; WT, weighted/weight.

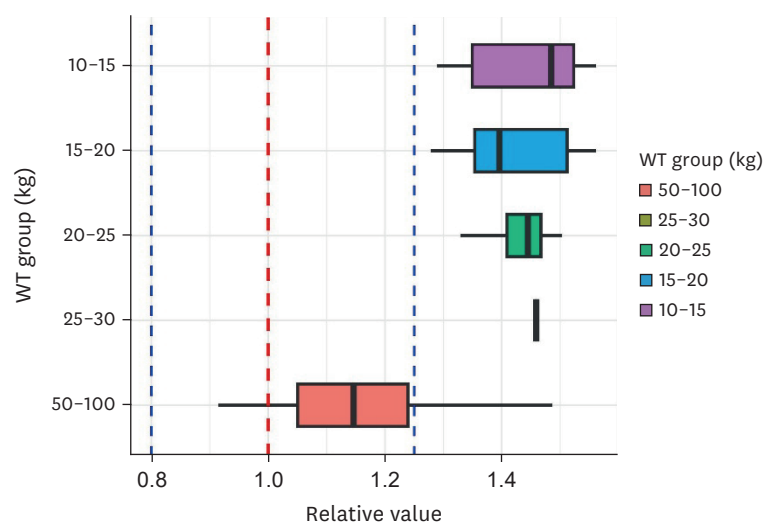


Figure 11. Simulated relative drug exposures across weight tiers under the proposed dosing regimen. Red-dashed line represents target efficacy threshold derived from *in vivo* studies. WT, weighted/weight.

Minimum Anticipated Biological Effect Level (MABEL) approach, and guide Phase 2 dose planning based on clinical data outcomes.

Key point: Leveraging PK/PD MS to translate non-clinical finding into an evidence-based dosing strategy, guiding dose selection from FIH trials through Phase 2.

Methods & results: (A) Non-clinical phase: Given the drug's characteristics, limited *in vivo* data was available; therefore, prior information from a comparator drug was used to guide PK/PD simulations. For PD, *in vitro* data enabled estimation of PD parameters, and PD parameter ratios with the comparator drug were applied to generate 2 sets of parameters

for the simulation. A MABEL approach was implemented using the lowest efficacious concentrations from *in vitro* data to ensure safety. Various simulations identified a dose range that achieved both PK and PD targets. Validation was also performed by comparing simulation results to the dosing regimen of the comparator drug.

(B) Clinical phase: A 2-compartment PK model was developed based on Phase 1 data from healthy subjects, with weight identified as a significant covariate impacting PK. For PDs, a direct response model utilizing an inhibitory sigmoid $E_{\max}$ relationship between drug concentration and the PD outcome was chosen. Simulations were conducted to determine optimal dosing regimens that achieve specific target levels, incorporating both induction and maintenance phases.

Outcomes: A starting dose was established through predictions in the non-clinical phase using MABEL approaches and enabled initiation of the Phase 1 clinical trial under the IND. Data from Phase 1 was subsequently obtained and used to determine the efficacious dose and optimize dosing strategy for Phase 2 clinical development which supported the initiation of Phase 2 study through an IND amendment. This case illustrates a cyclic model-informed development process in which models are built and continuously updated based on available data from nonclinical to early stage of clinical development, ultimately producing results with high clinical utility at each step.

Case study 3. Efficacy and safety dose prediction for GLP-1/2 agonist for diabetes mellitus and obesity patients using PK/PD MS

Goal: To suggest an efficacious and safe dose for an Fc fusion protein that targets GLP-1 and GLP-2 receptors in Phase 1 clinical trials based on PK modeling with nonclinical data, followed by using Phase 1 data to propose a recommended Phase 2 dose targeting diabetes and obesity based on PK/PD modeling.

Key point: A PK/PD model was established to provide model-informed evidence for efficacious dose selection of the GLP-1/2 dual agonist, utilizing key indicators such as HbA1c, glucose levels, and body weights to explain the exposure-response relationship for specific indications, including diabetes and obesity.

Methods & results: (A) Non-clinical data: A PK model for the fusion protein was developed using monkey data, as this species is most predictive of human PKs. Single-species allometry was applied to predict human PK parameters, and simulations based on these parameters were conducted to suggest a FIH dose, focusing on achieving target efficacious/safe exposure levels for Phase 1 clinical trials.

(B) Clinical data: A 2-compartment PK model was developed based on Phase 1 data from healthy subjects. For each indication, following PD models were developed.

- Diabetes mellitus: For the PD models, indirect response structures were applied to describe HbA1c and glucose responses. HbA1c was modeled using an inhibitory effect on the input rate, whereas post-oral glucose tolerance test (OGTT) glucose was described using a stimulatory effect on the output rate to reflect enhanced glucose lowering following treatment [28-30].
- Obesity: An indirect response model was used to describe body weight changes, with the drug effect modeled as inhibition of the weight gain input rate.

Simulations were performed for various dosing regimens, and Phase 2 dosing strategies were proposed to achieve the target levels. An additional Phase 1 cohort in obese patients is currently ongoing, which is expected to refine the PK/PD model and propose optimal dosing strategies for Phase 2 by incorporating covariates such as body weight that impact PK/PD.

Outcomes: Human PK predictions using one-species allometry using monkey data provided appropriate FIH dose ranges, which resulted in Phase 1 IND to become effective from the MFDS. Phase 1 results were subsequently obtained and are being leveraged to propose an optimal Phase 2 dose that meets efficacy criteria for both diabetes and obesity, supported by various simulations conducted under the Phase 2 dosing regimen (Figs. 12 and 13). This case

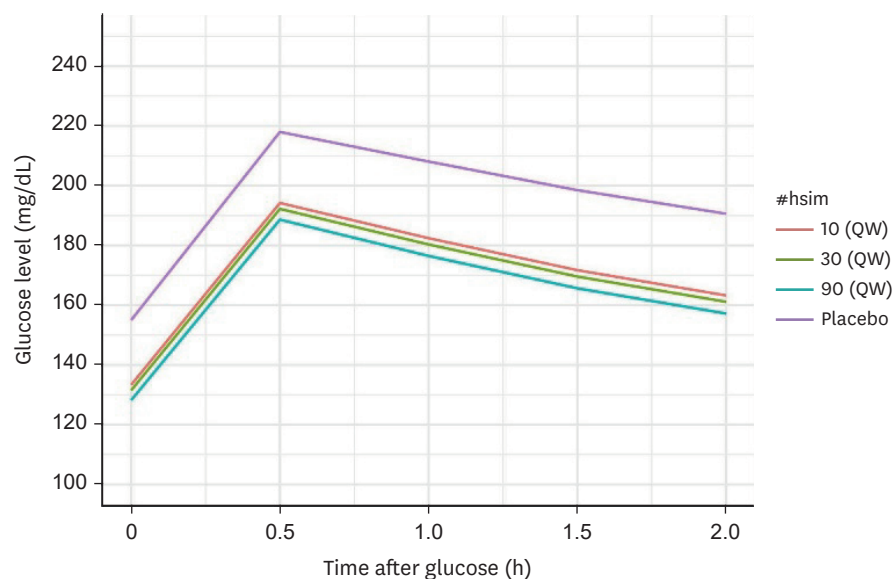


Figure 12. Simulated glucose level in oral glucose tolerance test under various dosing regimens (illustrative). QW, once weekly.

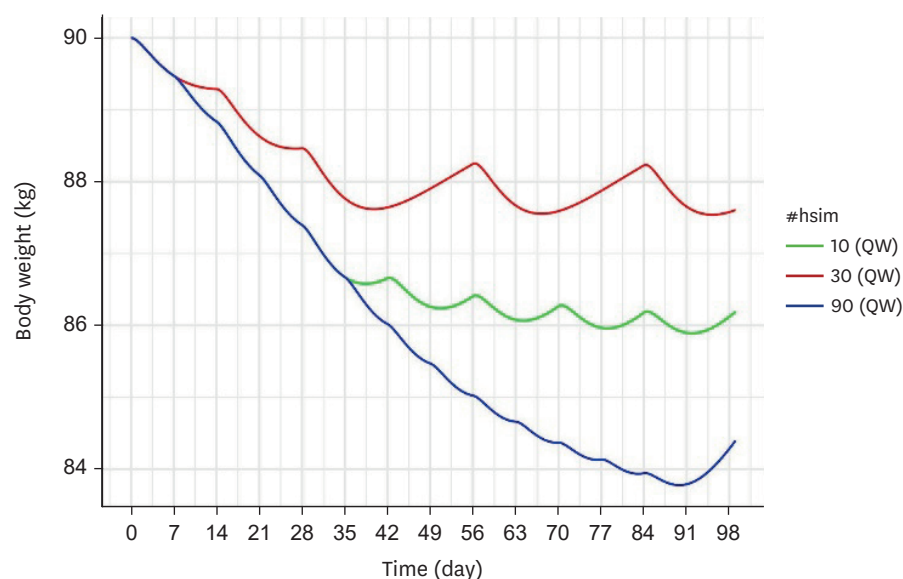


Figure 13. Simulated body weight changes under various dosing regimens (illustrative). QW, once weekly.

illustrates a cyclic model-informed development process in which Phase 1 clinical data were incorporated to develop multiple PD models for commonly evaluated GLP-1 agonist biomarkers—including glucose response (OGTT), HbA1c, and body weight—and to perform extensive simulations to identify optimal dosing regimens that satisfied predefined efficacy criteria while maintaining acceptable safety margins for each indication.

Case study 4. Evaluation of clinical DDI risk of an anticancer drug using PBPK modeling

Goal: To assess DDI risks of an oral anticancer drug co-administered with standard-of-care medications.

Key point: The evaluation was based on the *in vitro* metabolism and transporter data, along with available human clinical PK information. The potential for DDIs of the anticancer drug was assessed in accordance with the relevant U.S. FDA guidance for industry on drug interaction studies (2020) [31].

Methods & results: Initial assessment for DDI was performed using the basic model based on the *in vitro* inhibition data. The calculated R values of the anticancer drug exceeded the regulatory cutoff thresholds ($R_{i, \text{gut}} > 11$), suggesting that the anticancer drug had the potential to inhibit CYP3A4 *in vivo*. Additionally, the anticancer drug was confirmed as a substrate of CYP3A4 on *in vitro* metabolism studies.

These findings warranted further evaluation using PBPK modeling approaches. PBPK models were developed for the anticancer drug, midazolam (a sensitive substrate of CYP3A4), and probe perpetrators including itraconazole (CYP3A4 inhibitor) and rifampicin (strong CYP3A4 inducer). The modeling framework also included potential co-administered anticancer drugs, such as olaparib, gefitinib, and erlotinib.

For model verification and validation, the simulated PK exposures of each drug were compared to the available observed data. The simulated AUC and C_{max} ratios of substrates in the presence and absence of probe perpetrators were also compared where clinical observed values were available, followed by sensitivity analyses to assess parameter variability and model robustness. The simulated results showed that the AUC and C_{max} ratios of substrates remained within 1.20-fold (below the regulatory cutoff of 1.25), indicating that the potential clinical DDIs involving the anticancer drug were minimal.

Outcomes: The PBPK modeling approach supported the assessment of DDI risk for the investigational anticancer drug and helped guide the selection of appropriate co-administered medications. By providing insights into PK variability across various dosing regimens, metabolic enzyme interactions, and study populations, the PBPK modeling analysis informed internal drug development decision-making process and is expected to support future regulatory submissions.

Case study 5. PK/PD modeling for safe dosing and exposure prediction of nanoparticle anticancer agent

Goal: To predict safe dose ranges minimizing neutropenia risk in a nanoparticle-based anticancer agent, and to quantify the exposure contributions of nanoparticulated versus dissolved forms.

Key point: A multi-compartment PK/PD model enabled prediction of product and metabolite, and neutrophil dynamics across dose ranges and separation of nanoparticulated versus dissolved analyte profiles.

Methods & results: A 6-compartment PK model was developed with each analyte using a 2-compartment structure. For the PD model, a 4-transit compartment model was developed, incorporating feedback mechanisms where drug effects were modeled using $E_{\max}$ equation to describe effects on absolute neutrophil count (ANC).

ANC data were simulated for 6 weeks (3 cycles) after multiple intravenous infusions of the anticancer drug at various doses. Dose ranges were identified that minimized the risk of Grade III or IV neutropenia (**Fig. 14**).

Outcomes: The developed PK/PD models for anticancer drug effectively described the plasma and ANC dynamics of the nanoparticle-based agent. Simulations provided insights into safe dose ranges to avoid severe neutropenia, supporting the utility of nanoparticle-based drug delivery systems in improving therapeutic outcomes and minimizing adverse effects. The extended PK model further demonstrated that, compared with conventional anticancer therapy, nanoparticle-based delivery enables administration of higher doses while enhancing efficacy and securing a broader therapeutic window. This modeling results supported the Phase 1 IND becoming effective with the FDA.

These 5 case studies demonstrate how pharmacometrics modeling—ranging from PopPK and PK/PD to PBPK approaches—was strategically applied across therapeutic areas to inform dose selection, translate nonclinical data to humans, and support regulatory decisions and internal decision-making. By integrating translational best practices, including virtual simulations and biomarker-informed modeling, these projects exemplify the growing role of pharmacometrics in enabling scientifically sound, patient-specific, and globally aligned drug development strategies in Korea.

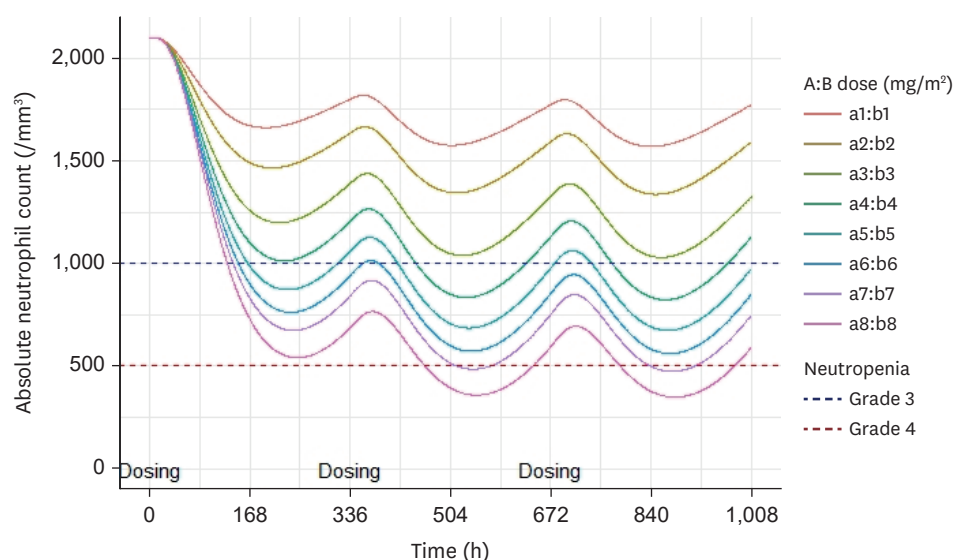


Figure 14. Simulated ANC profiles across dose levels of the nanoparticle-based formulation simulated time profiles of ANC across dose levels of the nanoparticle-based formulation. ANC, absolute neutrophil count.

CONCLUSION

The pharmacometrics in Korea stands at a pivotal moment, as the global transition toward AI/machine learning (ML)-enabled modeling and New Approach Methodologies (NAMs)—accelerated by recent regulatory initiatives such as the FDA’s 2025 NAMs roadmap—calls for adaptation, innovation, and strengthened scientific rigor [32,33]. This review, based on over a decade of modeling projects, highlights how Korea’s largest pharmacometrics consulting group has evolved in parallel with the shifting needs of drug development, supporting both regional and international programs.

The presented lists of emerging modality projects and the 5 case studies presented illustrate how model-informed strategies have delivered practical, regulatory-impactful solutions. These real-world applications demonstrate the critical role of pharmacometrics in enhancing decision-making, ensuring safety, and improving translational efficiency in Korean drug development.

Drawing from our project experiences, 5 core strategies emerged as essential for a successful pharmacometrics analysis program: clearly identifying the scientific question, interpreting regulatory expectations, maintaining transparent communication, delivering clinically meaningful results, and integrating advanced modeling methods with empirical approaches. These best practices, developed in response to Korea’s rapidly growing pipeline of biologics, biosimilars, and gene and cell therapies, offer actionable insights for the next generation of pharmacometricians and drug developers [34].

Because the dataset analyzed in this study was derived from projects conducted by a single service organization, the findings may not fully represent all pharmacometrics activities in Korea. However, given the organization’s collaborations with a broad range of regional pharmaceutical companies, academic institutions, and regulatory-related programs, the dataset likely reflects major trends in pharmacometrics applications during the study period.

As the field advances into a new paradigm of data-rich, AI-integrated, and globally harmonized drug development, the continued application of translational best practices—coupled with scientific flexibility and regulatory alignment—will remain essential for innovation. Future efforts should focus on strengthening system-level pharmacology modeling frameworks with validated predictive performance to support regulatory acceptance in Korea, while integrating AI/ML-driven modeling approaches to enhance translational relevance and efficiency in next-generation drug development [35,36].

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